1) a)

$$\frac{\mathrm{d}}{\mathrm{d}x}(\sinh x) = \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\mathrm{e}^{x}-\mathrm{e}^{-x}}{2}\right) = \frac{\mathrm{e}^{x}+\mathrm{e}^{-x}}{2} = \cosh x$$

1) b)

$$\frac{\mathrm{d}}{\mathrm{d}x}(\cosh x) = \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\mathrm{e}^{x}+\mathrm{e}^{-x}}{2}\right) = \frac{\mathrm{e}^{x}-\mathrm{e}^{-x}}{2} = \sinh x$$

1) c)

$$\frac{\mathrm{d}}{\mathrm{d}x}(\tanh x) = \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\mathrm{e}^{x}-\mathrm{e}^{-x}}{\mathrm{e}^{x}+\mathrm{e}^{-x}}\right)$$
$$= \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\mathrm{e}^{2x}-1}{\mathrm{e}^{2x}+1}\right)$$
$$= \frac{(2\mathrm{e}^{2x})(\mathrm{e}^{2x}+1)-(\mathrm{e}^{2x}-1)(2\mathrm{e}^{2x})}{(\mathrm{e}^{2x}+1)^{2}}$$
$$= \frac{(2\mathrm{e}^{4x}+2\mathrm{e}^{2x})-(2\mathrm{e}^{4x}-2\mathrm{e}^{2x})}{\mathrm{e}^{4x}+2\mathrm{e}^{2x}+1}$$
$$= \frac{4\mathrm{e}^{2x}}{\mathrm{e}^{4x}+2\mathrm{e}^{2x}+1}$$
$$= \frac{4}{\mathrm{e}^{2x}+2+\mathrm{e}^{-2x}}$$
$$= \frac{4}{(\mathrm{e}^{x}+\mathrm{e}^{-x})^{2}}$$
$$= \left(\frac{2}{\mathrm{e}^{x}+\mathrm{e}^{-x}}\right)^{2}$$
$$= \left(\frac{1}{\cosh x}\right)^{2}$$
$$= (\operatorname{sech} x)^{2} = \operatorname{sech}^{2} x$$

2) a)

Let $\mathrm{f}(x) = \sinh x$

$\Rightarrow \mathrm{f}(x+h) = \sinh(x+h)$

$\mathrm{f}'(x) = \lim_{h\to 0} \frac{\mathrm{f}(x+h)-\mathrm{f}(x)}{h}$

$= \lim_{h\to 0} \frac{\sinh(x+h)-\sinh x}{h}$

$= \lim_{h\to 0} \frac{\sinh x\cosh h+\cosh x\sinh h-\sinh x}{h}$

$= \lim_{h\to 0} \frac{\sinh x(\cosh h-1)+\cosh x(\sinh h)}{h}$

$= \lim_{h\to 0} \frac{\sinh x(\cosh h-1)}{h} + \lim_{h\to 0} \frac{\cosh x(\sinh h)}{h}$

$= \sinh x\left(\lim_{h\to 0} \frac{\cosh h-1}{h}\right) + \cosh x\left(\lim_{h\to 0} \frac{\sinh h}{h}\right)$

$= \sinh x \times 0 + \cosh x \times 1$

$= \cosh x$

2) b)

Let $\mathrm{f}(x) = \cosh x$

$\Rightarrow \mathrm{f}(x+h) = \cosh(x+h)$

$\mathrm{f}'(x) = \lim_{h\to 0} \frac{\mathrm{f}(x+h)-\mathrm{f}(x)}{h}$

$= \lim_{h\to 0} \frac{\cosh(x+h)-\cosh x}{h}$

$= \lim_{h\to 0} \frac{\cosh x\cosh h+\sinh x\sinh h-\cosh x}{h}$

$= \lim_{h\to 0} \frac{\cosh x(\cosh h-1)+\sinh x(\sinh h)}{h}$

$= \lim_{h\to 0} \frac{\cosh x(\cosh h-1)}{h} + \lim_{h\to 0} \frac{\sinh x(\sinh h)}{h}$

$= \cosh x\left(\lim_{h\to 0} \frac{\cosh h-1}{h}\right) + \sinh x\left(\lim_{h\to 0} \frac{\sinh h}{h}\right)$

$= \cosh x \times 0 + \sinh x \times 1$

$= \sinh x$

3) a)

$y = \sinh 2x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 2\cosh 2x$

3) b)

$y = \cosh 3x$

$\frac{dy}{dx} = 3\sinh 3x$

3) c)

$y = \tanh 6x$

$\frac{dy}{dx} = 6\,\mathrm{sech}^2\, 6x$

3) d)

$y = \sinh \frac{1}{4}x$

$\frac{dy}{dx} = \frac{1}{4}\cosh \frac{1}{4}x$

3) e)

$y = \cosh \frac{x}{3}$

$\frac{dy}{dx} = \frac{1}{3}\sinh \frac{x}{3}$

3) f)

$y = \tanh\left(-\frac{x}{5}\right)$

$\frac{dy}{dx} = -\frac{1}{5}\mathrm{sech}^2\left(-\frac{x}{5}\right)$

Alternatively:

$y = \tanh\left(-\frac{x}{5}\right) = -\tanh \frac{x}{5}$

$\frac{dy}{dx} = -\frac{1}{5}\mathrm{sech}^2 \frac{x}{5}$

4) a)

$y = 3\sinh 4x - 6\tanh 3x$

$\frac{dy}{dx} = 12\cosh 4x - 18\,\mathrm{sech}^2\, 3x$

4) b)

$y = 5\cosh \frac{x}{10} + \frac{1}{2}\tanh \frac{x}{4}$

$\frac{dy}{dx} = \frac{1}{2}\sinh \frac{x}{10} + \frac{1}{8}\mathrm{sech}^2 \frac{x}{4}$

5)

$y = 3\tanh\frac{1}{2}x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 3 \times \frac{1}{2}\operatorname{sech}^2\frac{1}{2}x$

$= \frac{3}{2}\operatorname{sech}^2\frac{1}{2}x$

$\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{x=\ln 16} = \frac{3}{2}\operatorname{sech}^2\left(\frac{1}{2}\ln 16\right)$

$= \frac{3}{2} \times \frac{1}{\left(\cosh\left(\frac{1}{2}\ln 16\right)\right)^2}$

$= \frac{3}{2} \times \frac{1}{\left(\cosh\left(\ln 16^{\frac{1}{2}}\right)\right)^2}$

$= \frac{3}{2} \times \frac{1}{(\cosh(\ln 4))^2}$

$= \frac{3}{2} \times \frac{1}{\left(\frac{\mathrm{e}^{\ln 4}+\mathrm{e}^{-\ln 4}}{2}\right)^2}$

$= \frac{3}{2} \times \frac{1}{\left(\frac{\mathrm{e}^{\ln 4}+\mathrm{e}^{\ln 4^{-1}}}{2}\right)^2}$

$= \frac{3}{2} \times \frac{1}{\left(\frac{\mathrm{e}^{\ln 4}+\mathrm{e}^{\ln\frac{1}{4}}}{2}\right)^2}$

$= \frac{3}{2} \times \frac{1}{\left(\frac{4+\frac{1}{4}}{2}\right)^2}$

$= \frac{96}{289}$

6)

$y = 3\sinh x$

$y|_{x=\ln 6} = 3 \times \frac{e^{\ln 6} - e^{-\ln 6}}{2}$

$= \frac{3}{2}\left(e^{\ln 6} - e^{\ln \frac{1}{6}}\right)$

$= \frac{3}{2}\left(6 - \frac{1}{6}\right) = \frac{35}{4}$

$\frac{dy}{dx} = 3\cosh x$

$\left.\frac{dy}{dx}\right|_{x=\ln 6} = 3 \times \frac{e^{\ln 6} + e^{-\ln 6}}{2}$

$= \frac{3}{2}\left(e^{\ln 6} + e^{\ln \frac{1}{6}}\right)$

$= \frac{3}{2}\left(6 + \frac{1}{6}\right) = \frac{37}{4}$

So the tangent will have equation $y - \frac{35}{4} = \frac{37}{4}(x - \ln 6)$

$\Rightarrow 4y - 35 = 37x - 37\ln 6$

$\Rightarrow -37x + 4y = 35 - 37\ln 6$ or $37x - 4y = 37\ln 6 - 35$

7)

$y = 5\sinh x - \cosh x$

$y|_{x=\ln 2} = 5 \times \frac{e^{\ln 2} - e^{-\ln 2}}{2} - \frac{e^{\ln 2} + e^{-\ln 2}}{2}$

$= \frac{5}{2}\left(e^{\ln 2} - e^{\ln \frac{1}{2}}\right) - \frac{1}{2}\left(e^{\ln 2} + e^{\ln \frac{1}{2}}\right)$

$= \frac{5}{2}\left(2 - \frac{1}{2}\right) - \frac{1}{2}\left(2 + \frac{1}{2}\right) = \frac{5}{2}$

$\frac{dy}{dx} = 5\cosh x - \sinh x$

$\left.\frac{dy}{dx}\right|_{x=\ln 2} = 5 \times \frac{e^{\ln 2} + e^{-\ln 2}}{2} - \frac{e^{\ln 2} - e^{-\ln 2}}{2}$

$= \frac{5}{2}\left(e^{\ln 2} + e^{\ln \frac{1}{2}}\right) - \frac{1}{2}\left(e^{\ln 2} - e^{\ln \frac{1}{2}}\right)$

$= \frac{5}{2}\left(2 + \frac{1}{2}\right) - \frac{1}{2}\left(2 - \frac{1}{2}\right) = \frac{11}{2}$

So the tangent will have equation $y - \frac{5}{2} = \frac{11}{2}(x - \ln 2)$

$\Rightarrow y = \frac{11}{2}x + \frac{5}{2} - \frac{11}{2}\ln 2$

8)

$y = 3\cosh 2x - 4\sinh 3x$

$y|_{x=\ln\frac{1}{3}} = 3 \times \frac{e^{2\ln\frac{1}{3}} + e^{-2\ln\frac{1}{3}}}{2} - 4 \times \frac{e^{3\ln\frac{1}{3}} - e^{-3\ln\frac{1}{3}}}{2}$

$= \frac{3}{2}\left(e^{\ln\frac{1}{9}} + e^{\ln 9}\right) - 2\left(e^{\ln\frac{1}{27}} - e^{\ln 27}\right)$

$= \frac{3}{2}\left(\frac{1}{9} + 9\right) - 2\left(\frac{1}{27} - 27\right) = \frac{1825}{27}$

$\frac{dy}{dx} = 6\sinh 2x - 12\cosh 3x$

$\left.\frac{dy}{dx}\right|_{x=\ln\frac{1}{3}} = 6 \times \frac{e^{2\ln\frac{1}{3}} - e^{-2\ln\frac{1}{3}}}{2} - 12 \times \frac{e^{3\ln\frac{1}{3}} + e^{-3\ln\frac{1}{3}}}{2}$

$= 3\left(e^{\ln\frac{1}{9}} - e^{\ln 9}\right) - 6\left(e^{\ln\frac{1}{27}} + e^{\ln 27}\right)$

$= 3\left(\frac{1}{9} - 9\right) - 6\left(\frac{1}{27} + 27\right) = -\frac{1700}{9}$

So the tangent will have equation $y - \frac{1825}{27} = -\frac{1700}{9}\left(x - \ln\frac{1}{3}\right)$

$\Rightarrow y = -\frac{1700}{9}x + \frac{1825}{27} + \frac{1700}{9}\ln\frac{1}{3}$

9)

$y = \tanh\frac{x}{2}$

$y|_{x=\ln 4} = \frac{e^{\frac{1}{2}\ln 4} - e^{-\frac{1}{2}\ln 4}}{e^{\frac{1}{2}\ln 4} + e^{-\frac{1}{2}\ln 4}}$

$= \frac{e^{\ln 2} - e^{\ln\frac{1}{2}}}{e^{\ln 2} + e^{\ln\frac{1}{2}}}$

$= \frac{2 - \frac{1}{2}}{2 + \frac{1}{2}} = \frac{3}{5}$

$\frac{dy}{dx} = \frac{1}{2}\operatorname{sech}^2\frac{x}{2}$

$\left.\frac{dy}{dx}\right|_{x=\ln 4} = \frac{1}{2} \times \frac{1}{\cosh^2\frac{x}{2}}$

$= \frac{1}{2} \times \frac{1}{\left(\frac{e^{\frac{1}{2}\ln 4} + e^{-\frac{1}{2}\ln 4}}{2}\right)^2}$

$= \frac{1}{2} \times \left(\frac{2}{e^{\frac{1}{2}\ln 4} + e^{-\frac{1}{2}\ln 4}}\right)^2$

$= \frac{1}{2} \times \left(\frac{2}{e^{\ln 2} + e^{\ln\frac{1}{2}}}\right)^2$

$= \frac{1}{2} \times \left(\frac{2}{2 + \frac{1}{2}}\right)^2 = \frac{8}{25}$

So the tangent will have equation $y - \frac{3}{5} = \frac{8}{25}(x - \ln 4)$

$\Rightarrow y = \frac{8}{25}x + \frac{3}{5} - \frac{8}{25}\ln 4$

10)

$y = 8 + 6\tanh\frac{1}{3}x - 4\tanh\frac{2}{3}x$

$y|_{x=0} = 8 + 6\tanh 0 - 4\tanh 0 = 8$

$\frac{dy}{dx} = 6 \times \frac{1}{3}\operatorname{sech}^2\frac{1}{3}x - 4 \times \frac{2}{3}\operatorname{sech}^2\frac{2}{3}x$

$= 2\operatorname{sech}^2\frac{1}{3}x - \frac{8}{3}\operatorname{sech}^2\frac{2}{3}x$

$\left.\frac{dy}{dx}\right|_{x=0} = 2\operatorname{sech}^2 0 - \frac{8}{3}\operatorname{sech}^2 0$

$= \frac{2}{(\cosh 0)^2} - \frac{8}{3(\cosh 0)^2}$

$= \frac{2}{1} - \frac{8}{3} = -\frac{2}{3}$

So the normal will have equation $y - 8 = \frac{3}{2}(x - 0)$

$\Rightarrow y = \frac{3}{2}x + 8$

11)

First find l_1 :

$y = 12\sinh\frac{x}{3}$

$y|_{x=\ln 8} = 12\sinh\left(\frac{1}{3}\ln 8\right)$

$= 12\sinh\left(\ln 8^{\frac{1}{3}}\right)$

$= 12\sinh(\ln 2)$

$= 12 \times \frac{e^{\ln 2} - e^{-\ln 2}}{2}$

$= 12 \times \frac{e^{\ln 2} - e^{\ln 2^{-1}}}{2}$

$= 12 \times \frac{e^{\ln 2} - e^{\ln \frac{1}{2}}}{2}$

$= 12 \times \frac{2 - \frac{1}{2}}{2} = 9$

$\frac{dy}{dx} = 12 \times \frac{1}{3}\cosh\frac{x}{3}$

$= 4\cosh\frac{x}{3}$

$\left.\frac{dy}{dx}\right|_{x=\ln 8} = 4\cosh\left(\frac{1}{3}\ln 8\right)$

$= 4\cosh\left(\ln 8^{\frac{1}{3}}\right)$

$= 4\cosh(\ln 2)$

$= 4 \times \frac{e^{\ln 2} + e^{-\ln 2}}{2}$

$= 4 \times \frac{e^{\ln 2} + e^{\ln 2^{-1}}}{2}$

$= 4 \times \frac{e^{\ln 2} + e^{\ln \frac{1}{2}}}{2}$

$= 4 \times \frac{2 + \frac{1}{2}}{2} = 5$

So l_1 has equation $y - 9 = -\frac{1}{5}(x - \ln 8)$

$\Rightarrow y = -\frac{1}{5}x + \frac{1}{5}\ln 8 + 9$

Now find l_2 :

$y = 6\cosh\frac{x}{4}$

$y|_{x=\ln 16} = 6\cosh\left(\frac{1}{4}\ln 16\right)$

$= 6\cosh\left(\ln 16^{\frac{1}{4}}\right)$

$= 6\cosh(\ln 2)$

$= 6 \times \frac{e^{\ln 2}+e^{-\ln 2}}{2}$

$= 6 \times \frac{e^{\ln 2}+e^{\ln 2^{-1}}}{2}$

$= 6 \times \frac{e^{\ln 2}+e^{\ln \frac{1}{2}}}{2}$

$= 6 \times \frac{2+\frac{1}{2}}{2} = \frac{15}{2}$

$\frac{dy}{dx} = 6 \times \frac{1}{4}\sinh\frac{x}{4}$

$= \frac{3}{2}\sinh\frac{x}{4}$

$\left.\frac{dy}{dx}\right|_{x=\ln 16} = \frac{3}{2}\sinh\left(\frac{1}{4}\ln 16\right)$

$= \frac{3}{2}\sinh\left(\ln 16^{\frac{1}{4}}\right)$

$= \frac{3}{2}\sinh(\ln 2)$

$= \frac{3}{2} \times \frac{e^{\ln 2}-e^{-\ln 2}}{2}$

$= \frac{3}{2} \times \frac{e^{\ln 2}-e^{\ln 2^{-1}}}{2}$

$= \frac{3}{2} \times \frac{e^{\ln 2}-e^{\ln \frac{1}{2}}}{2}$

$= \frac{3}{2} \times \frac{2-\frac{1}{2}}{2} = \frac{9}{8}$

So l_2 has equation $y - \frac{15}{2} = -\frac{8}{9}(x - \ln 16)$

$\Rightarrow y = -\frac{8}{9}x + \frac{8}{9}\ln 16 + \frac{15}{2}$

l_1 and l_2 intersect when:

$-\frac{1}{5}x + \frac{1}{5}\ln 8 + 9 = -\frac{8}{9}x + \frac{8}{9}\ln 16 + \frac{15}{2}$

$\Rightarrow \frac{31}{45}x = \frac{8}{9}\ln 16 - \frac{1}{5}\ln 8 - \frac{3}{2}$

$\Rightarrow \frac{31}{45}x = \frac{8}{9}\ln 2^4 - \frac{1}{5}\ln 2^3 - \frac{3}{2}$

$\Rightarrow \frac{31}{45}x = \frac{32}{9}\ln 2 - \frac{3}{5}\ln 2 - \frac{3}{2}$

$\Rightarrow \frac{31}{45}x = \frac{133}{45}\ln 2 - \frac{3}{2}$

$\Rightarrow x = \frac{133}{31}\ln 2 - \frac{135}{62}$

12)

$y = \sinh x + \tanh x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \cosh x + \operatorname{sech}^2 x$

$\cosh x \geq 1$ for all real values of x

So as $\operatorname{sech}^2 x = \frac{1}{\cosh^2 x}$, it must be that $0 < \operatorname{sech}^2 x \leq 1$

So $\cosh x + \operatorname{sech}^2 x > 1$

$\therefore \frac{\mathrm{d}y}{\mathrm{d}x} > 0$ for all x, and hence the curve is always increasing.

13)

$y = \sinh x + 2\cosh x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \cosh x + 2\sinh x$

Stationary points exist when $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$

$\therefore \cosh x + 2\sinh x = 0$

$\Rightarrow 1 + 2\tanh x = 0$

$\Rightarrow \tanh x = -\frac{1}{2}$

$\Rightarrow x = \frac{1}{2}\ln\left(\frac{1+\left(-\frac{1}{2}\right)}{1-\left(-\frac{1}{2}\right)}\right) = \frac{1}{2}\ln\frac{1}{3} = \ln\frac{1}{\sqrt{3}}$

$y|_{x=\frac{1}{2}\ln\frac{1}{3}} = \sinh\left(\ln\frac{1}{\sqrt{3}}\right) + 2\cosh\left(\ln\frac{1}{\sqrt{3}}\right)$

$= \frac{\mathrm{e}^{\ln\frac{1}{\sqrt{3}}} - \mathrm{e}^{-\ln\frac{1}{\sqrt{3}}}}{2} + 2 \times \frac{\mathrm{e}^{\ln\frac{1}{\sqrt{3}}} + \mathrm{e}^{-\ln\frac{1}{\sqrt{3}}}}{2}$

$= \frac{\mathrm{e}^{\ln\frac{1}{\sqrt{3}}} - \mathrm{e}^{\ln\sqrt{3}}}{2} + 2 \times \frac{\mathrm{e}^{\ln\frac{1}{\sqrt{3}}} + \mathrm{e}^{\ln\sqrt{3}}}{2}$

$= \frac{\frac{1}{\sqrt{3}} - \sqrt{3}}{2} + 2 \times \frac{\frac{1}{\sqrt{3}} + \sqrt{3}}{2}$

$= \frac{1}{2\sqrt{3}} - \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{3}} + \sqrt{3}$

$= \frac{\sqrt{3}}{6} - \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{3} + \sqrt{3}$

$= \sqrt{3}$

So the stationary point is at $\left(\ln\frac{1}{\sqrt{3}}, \sqrt{3}\right)$

14)

$y = 9\cosh x + 3\sinh 2x$

$\frac{dy}{dx} = 9\sinh x + 6\cosh 2x$

$= 9\sinh x + 6(1 + 2\sinh^2 x)$

$= 12\sinh^2 x + 9\sinh x + 6$

$\frac{dy}{dx} = 1$

$\Rightarrow 12\sinh^2 x + 9\sinh x + 6 = 1$

$\Rightarrow 12\sinh^2 x + 9\sinh x + 5 = 0$

Discriminant: $b^2 - 4ac = 9^2 - 4 \times 12 \times 5 = -159 < 0$

The quadratic has no real roots, hence there is no real value of $\sinh x$ such that the equation is zero (and therefore no real x).

So there is no point on the curve where the gradient is 1

15)

$y = p\sinh x - q\cosh x$

$\frac{dy}{dx} = p\cosh x - q\sinh x$

Stationary points exist when $\frac{dy}{dx} = 0$

$\therefore p\cosh x - q\sinh x = 0$

$\Rightarrow p - \frac{q\sinh x}{\cosh x} = 0$

$\Rightarrow p - q\tanh x = 0$

$\Rightarrow \tanh x = \frac{p}{q}$

If $p \geq q \Rightarrow \frac{p}{q} \geq 1$

However, $-1 < \tanh x < 1$, so $\tanh x = \frac{p}{q}$ has no real solutions

Hence the curve $y = p\sinh x - q\cosh x$ has no stationary points.

16)

$y = \sinh x - 8\tanh x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \cosh x - 8\,\mathrm{sech}^2 x$

Stationary points exist when $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$

$\therefore \cosh x - 8\,\mathrm{sech}^2 x = 0$

$\Rightarrow \cosh x - \frac{8}{\cosh^2 x} = 0$

$\Rightarrow \cosh^3 x - 8 = 0$

$\Rightarrow \cosh^3 x = 8$

$\Rightarrow \cosh x = 2$

$\Rightarrow x = \pm\ln\left(2 + \sqrt{2^2 - 1}\right) = \pm\ln\left(2 + \sqrt{3}\right) = \ln\left(2 \pm \sqrt{3}\right)$

$y|_{x=\ln(2+\sqrt{3})} = \sinh\left(\ln\left(2 + \sqrt{3}\right)\right) - 8\tanh\left(\ln\left(2 + \sqrt{3}\right)\right)$

$= \frac{e^{\ln(2+\sqrt{3})} - e^{-\ln(2+\sqrt{3})}}{2} - 8 \times \frac{e^{\ln(2+\sqrt{3})} - e^{-\ln(2+\sqrt{3})}}{e^{\ln(2+\sqrt{3})} + e^{-\ln(2+\sqrt{3})}}$

$= \frac{e^{\ln(2+\sqrt{3})} - e^{\ln(2+\sqrt{3})^{-1}}}{2} - 8 \times \frac{e^{\ln(2+\sqrt{3})} - e^{\ln(2+\sqrt{3})^{-1}}}{e^{\ln(2+\sqrt{3})} + e^{\ln(2+\sqrt{3})^{-1}}}$

$= \frac{e^{\ln(2+\sqrt{3})} - e^{\ln(2-\sqrt{3})}}{2} - 8 \times \frac{e^{\ln(2+\sqrt{3})} - e^{\ln(2-\sqrt{3})}}{e^{\ln(2+\sqrt{3})} + e^{\ln(2-\sqrt{3})}}$

$= \frac{(2+\sqrt{3}) - (2-\sqrt{3})}{2} - 8 \times \frac{(2+\sqrt{3}) - (2-\sqrt{3})}{(2+\sqrt{3}) + (2-\sqrt{3})}$

$= \frac{2\sqrt{3}}{2} - 8 \times \frac{2\sqrt{3}}{4}$

$= \sqrt{3} - 4\sqrt{3}$

$= -3\sqrt{3}$

So there is a stationary point is at $\left(\ln\left(2 + \sqrt{3}\right), -3\sqrt{3}\right)$

$y|_{x=\ln(2-\sqrt{3})} = \sinh\left(\ln\left(2 - \sqrt{3}\right)\right) - 8\tanh\left(\ln\left(2 - \sqrt{3}\right)\right)$

$= \frac{e^{\ln(2-\sqrt{3})} - e^{-\ln(2-\sqrt{3})}}{2} - 8 \times \frac{e^{\ln(2-\sqrt{3})} - e^{-\ln(2-\sqrt{3})}}{e^{\ln(2-\sqrt{3})} + e^{-\ln(2-\sqrt{3})}}$

$= \frac{e^{\ln(2-\sqrt{3})} - e^{\ln(2-\sqrt{3})^{-1}}}{2} - 8 \times \frac{e^{\ln(2-\sqrt{3})} - e^{\ln(2-\sqrt{3})^{-1}}}{e^{\ln(2-\sqrt{3})} + e^{\ln(2-\sqrt{3})^{-1}}}$

$= \frac{e^{\ln(2-\sqrt{3})} - e^{\ln(2+\sqrt{3})}}{2} - 8 \times \frac{e^{\ln(2-\sqrt{3})} - e^{\ln(2+\sqrt{3})}}{e^{\ln(2-\sqrt{3})} + e^{\ln(2+\sqrt{3})}}$

$= \frac{(2-\sqrt{3}) - (2+\sqrt{3})}{2} - 8 \times \frac{(2-\sqrt{3}) - (2+\sqrt{3})}{(2-\sqrt{3}) + (2+\sqrt{3})}$

$= \frac{-2\sqrt{3}}{2} - 8 \times \frac{-2\sqrt{3}}{4}$

$= -\sqrt{3} + 4\sqrt{3}$

$= 3\sqrt{3}$

So there is a stationary point is at $\left(\ln\left(2-\sqrt{3}\right), 3\sqrt{3}\right)$

So there are two stationary points, one at $\left(\ln\left(2+\sqrt{3}\right), -3\sqrt{3}\right)$ and the other at $\left(\ln\left(2-\sqrt{3}\right), 3\sqrt{3}\right)$

17)

$y = 5\cosh x + \sinh x + x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 5\sinh x + \cosh x + 1$

Stationary points exist when $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$

$\therefore 5\sinh x + \cosh x + 1 = 0$

$\Rightarrow 5 \times \frac{\mathrm{e}^{x}-\mathrm{e}^{-x}}{2} + \frac{\mathrm{e}^{x}+\mathrm{e}^{-x}}{2} + 1 = 0$

$\Rightarrow \frac{5}{2}\mathrm{e}^{x} - \frac{5}{2}\mathrm{e}^{-x} + \frac{1}{2}\mathrm{e}^{x} + \frac{1}{2}\mathrm{e}^{-x} + 1 = 0$

$\Rightarrow 3\mathrm{e}^{x} - 2\mathrm{e}^{-x} + 1 = 0$

$\Rightarrow 3\mathrm{e}^{2x} - 2 + \mathrm{e}^{x} = 0$

$\Rightarrow 3\mathrm{e}^{2x} + \mathrm{e}^{x} - 2 = 0$

$\Rightarrow (3\mathrm{e}^{x} - 2)(\mathrm{e}^{x} + 1) = 0$

$\Rightarrow \mathrm{e}^{x} = \frac{2}{3}$ or $\mathrm{e}^{x} = -1$

$\mathrm{e}^{x} \neq -1$ so no real solutions

If $\mathrm{e}^{x} = \frac{2}{3}$, then $x = \ln\frac{2}{3}$

So there is a single stationary point when $x = \ln\frac{2}{3}$

18)

$y = \cosh 3x - 2\sinh x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 3\sinh 3x - 2\cosh x$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 9\cosh 3x - 2\sinh x$

$\left.\frac{\mathrm{d}^2y}{\mathrm{d}x^2}\right|_{x=\ln 2} = 9\cosh(3\ln 2) - 2\sinh(\ln 2)$

$= 9\cosh(\ln 8) - 2\sinh(\ln 2)$

$= 9 \times \frac{\mathrm{e}^{\ln 8} + \mathrm{e}^{-\ln 8}}{2} - 2 \times \frac{\mathrm{e}^{\ln 2} - \mathrm{e}^{-\ln 2}}{2}$

$= 9 \times \frac{\mathrm{e}^{\ln 8} + \mathrm{e}^{\ln \frac{1}{8}}}{2} - 2 \times \frac{\mathrm{e}^{\ln 2} - \mathrm{e}^{\ln \frac{1}{2}}}{2}$

$= 9 \times \frac{8 + \frac{1}{8}}{2} - 2 \times \frac{2 - \frac{1}{2}}{2}$

$= \frac{585}{16} - \frac{3}{2} = \frac{561}{16} > 0$

As $\left.\frac{\mathrm{d}^2y}{\mathrm{d}x^2}\right|_{x=\ln 2} > 0$ the curve is convex (concave upwards) at $x = \ln 2$

19)

$y = 4\sinh 2x - 3\cosh 2x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 4 \times 2\cosh 2x - 3 \times 2\sinh 2x$

$= 8\cosh 2x - 6\sinh 2x$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 8 \times 2\sinh 2x - 6 \times 2\cosh 2x$

$= 16\sinh 2x - 12\cosh 2x$

Points of inflection exist when $\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 0$

$\therefore 16\sinh 2x - 12\cosh 2x = 0$

$\Rightarrow 16\sinh 2x = 12\cosh 2x$

$\Rightarrow \frac{\sinh 2x}{\cosh 2x} = \frac{12}{16}$

$\Rightarrow \tanh 2x = \frac{3}{4}$

$\Rightarrow 2x = \operatorname{artanh}\frac{3}{4}$

$\Rightarrow x = \frac{1}{2} \times \frac{1}{2}\ln\left(\frac{1+\frac{3}{4}}{1-\frac{3}{4}}\right) = \frac{1}{4}\ln 7$

$y|_{x=\frac{1}{4}\ln 7} = 4\sinh\left(2 \times \frac{1}{4}\ln 7\right) - 3\cosh\left(2 \times \frac{1}{4}\ln 7\right)$

$= 4\sinh\left(\frac{1}{2}\ln 7\right) - 3\cosh\left(\frac{1}{2}\ln 7\right)$

$= 4\sinh\left(\ln 7^{\frac{1}{2}}\right) - 3\cosh\left(\ln 7^{\frac{1}{2}}\right)$

$= 4 \times \frac{e^{\ln 7^{\frac{1}{2}}} - e^{-\ln 7^{\frac{1}{2}}}}{2} - 3 \times \frac{e^{\ln 7^{\frac{1}{2}}} + e^{-\ln 7^{\frac{1}{2}}}}{2}$

$= 4 \times \frac{e^{\ln 7^{\frac{1}{2}}} - e^{\ln 7^{-\frac{1}{2}}}}{2} - 3 \times \frac{e^{\ln 7^{\frac{1}{2}}} + e^{\ln 7^{-\frac{1}{2}}}}{2}$

$= 4 \times \frac{7^{\frac{1}{2}} - 7^{-\frac{1}{2}}}{2} - 3 \times \frac{7^{\frac{1}{2}} + 7^{-\frac{1}{2}}}{2}$

$= 2\left(\sqrt{7} - \frac{1}{\sqrt{7}}\right) - \frac{3}{2}\left(\sqrt{7} + \frac{1}{\sqrt{7}}\right)$

$= 2\left(\sqrt{7} - \frac{\sqrt{7}}{7}\right) - \frac{3}{2}\left(\sqrt{7} + \frac{\sqrt{7}}{7}\right)$

$= \frac{12\sqrt{7}}{7} - \frac{12\sqrt{7}}{7} = 0$

We need to check that the 2nd derivative changes sign either side of the point we have found. $\frac{1}{4}\ln 7 = 0.486$ (3 SF)

$\left.\frac{d^2y}{dx^2}\right|_{x=0} = 16\sinh 0 - 12\cosh 0 = -12 < 0$ (concave (concave downwards))

$\left.\frac{d^2y}{dx^2}\right|_{x=1} = 16\sinh 1 - 12\cosh 1 = 0.286\ldots > 0$ (convex (concave upwards))

$\therefore \left(\frac{1}{4}\ln 7, 0\right)$ is a point of inflection for the curve.

20) a)

$y = \sinh\left(\sqrt{x}\right) = \sinh\left(x^{\frac{1}{2}}\right)$

$\frac{dy}{dx} = \frac{1}{2}x^{-\frac{1}{2}}\cosh\left(x^{\frac{1}{2}}\right)$

20) b)

$y = \cosh\left(\frac{1}{\sqrt{x}}\right) = \cosh\left(x^{-\frac{1}{2}}\right)$

$\frac{dy}{dx} = -\frac{1}{2}x^{-\frac{3}{2}}\sinh\left(x^{-\frac{1}{2}}\right)$

20) c)

$y = \tanh\left(\frac{x+1}{2\sqrt{x}}\right) = \tanh\left(\frac{x}{2x^{\frac{1}{2}}} + \frac{1}{2x^{\frac{1}{2}}}\right) = \tanh\left(\frac{1}{2}x^{\frac{1}{2}} + \frac{1}{2}x^{-\frac{1}{2}}\right)$

$\frac{dy}{dx} = \left(\frac{1}{4}x^{-\frac{1}{2}} - \frac{1}{4}x^{-\frac{3}{2}}\right)\operatorname{sech}^2\left(\frac{x+1}{2\sqrt{x}}\right)$

21) a)

$y = \sinh^2 x = (\sinh x)^2$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 2\cosh x\,(\sinh x)^1 = 2\sinh x\cosh x$

21) b)

$y = \cosh^2 x = (\cosh x)^2$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 2\sinh x\,(\cosh x)^1 = 2\sinh x\cosh x$

21) c)

$y = \sinh^3 x = (\sinh x)^3$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 3\cosh x\,(\sinh x)^2 = 3\sinh^2 x\cosh x$

21) d)

$y = \cosh^4 x = (\cosh x)^4$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 4\sinh x\,(\cosh x)^3 = 4\sinh x\cosh^3 x$

21) e)

$y = \tanh^2 x = (\tanh x)^2$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 2\,\mathrm{sech}^2 x\,(\tanh x)^1 = 2\,\mathrm{sech}^2 x\tanh x$

21) f)

$y = \tanh^5 x = (\tanh x)^5$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 5\,\mathrm{sech}^2 x\,(\tanh x)^4 = 5\,\mathrm{sech}^2 x\tanh^4 x$

22)

$y = 3\cosh^2 x - \sinh x = 3(\cosh x)^2 - \sinh x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 3 \times 2\sinh x\,(\cosh x)^1 - \cosh x$

$= 6\sinh x \cosh x - \cosh x$

Stationary points exist when $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$

$\therefore 6\sinh x \cosh x - \cosh x = 0$

$\Rightarrow \cosh x\,(6\sinh x - 1) = 0$

$\cosh x = 0$ has no real solutions

$6\sinh x - 1 = 0 \Rightarrow \sinh x = \frac{1}{6}$

$\Rightarrow x = \ln\left(\frac{1}{6} + \sqrt{\left(\frac{1}{6}\right)^2 + 1}\right) = \ln\left(\frac{1}{6} + \frac{\sqrt{37}}{6}\right)$

So there is a single stationary point at $x = \ln\left(\frac{1}{6} + \frac{\sqrt{37}}{6}\right)$

23)

$\mathrm{f}(x) = \cosh^2 2x - p\cosh 2x = (\cosh 2x)^2 - p\cosh 2x$

$\mathrm{f}'(x) = 2\sinh 2x \times 2(\cosh 2x)^1 - 2p\sinh 2x$

$= 4\sinh 2x \cosh 2x - 2p\sinh 2x$

Stationary points exist when $\mathrm{f}'(x) = 0$

$4\sinh 2x \cosh 2x - 2p\sinh 2x = 0$

$\Rightarrow \sinh 2x\,(2\cosh 2x - p) = 0$

$\sinh 2x = 0 \Rightarrow x = 0$

So the curve has a stationary point at $x = 0$ for all real values of p

$2\cosh 2x - p = 0 \Rightarrow 2\cosh 2x = p$

This has real solutions if $p \geq 2$ as $\cosh 2x \geq 1$ for all real values of x

So the minimum value of p for the curve to have more than one stationary point is $p = 2$

24) a)

$y = \sinh(\cosh x)$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \sinh x \cosh(\cosh x)$

24) b)

$y = \cosh(\sinh x)$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \cosh x \sinh(\sinh x)$

24) c)

$y = \tanh(\tanh x)$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \operatorname{sech}^2 x \operatorname{sech}^2(\tanh x)$

25) a)

$\frac{\mathrm{d}}{\mathrm{d}x}(\ln(\sinh x)) = \frac{\cosh x}{\sinh x} = \coth x$

25) b)

$\frac{\mathrm{d}}{\mathrm{d}x}(\ln(\cosh x)) = \frac{\sinh x}{\cosh x} = \tanh x$

26)

$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{2}\ln(\tanh x)\right)$

$= \frac{1}{2} \times \frac{\operatorname{sech}^2 x}{\tanh x}$

$= \frac{1}{2} \times \frac{\frac{1}{\cosh^2 x}}{\frac{\sinh x}{\cosh x}}$

$= \frac{1}{2} \times \frac{1}{\cosh^2 x} \times \frac{\cosh x}{\sinh x}$

$= \frac{1}{2} \times \frac{1}{\sinh x \cosh x}$

$= \frac{1}{2 \sinh x \cosh x}$

$= \frac{1}{\sinh 2x}$

27) a)

$y = \sinh x \cosh x$

$\frac{dy}{dx} = \sinh x \times \sinh x + \cosh x \times \cosh x$

$= \sinh^2 x + \cosh^2 x$

Alternatively:

$y = \sinh x \cosh x = \frac{1}{2}\sinh 2x$

$\frac{dy}{dx} = \cosh 2x$

27) b)

$y = \sinh 3x \cosh 4x$

$\frac{dy}{dx} = \sinh 3x \times 4\sinh 4x + 3\cosh 3x \times \cosh 4x$

$= 4\sinh 3x \sinh 4x + 3\cosh 3x \cosh 4x$

27) c)

$y = x \sinh x$

$\frac{dy}{dx} = x \times \cosh x + 1 \times \sinh x$

$= x\cosh x + \sinh x$

27) d)

$y = x^2 \cosh x$

$\frac{dy}{dx} = x^2 \times \sinh x + 2x \times \cosh x$

$= x^2 \sinh x + 2x\cosh x$

27) e)

$y = (x+1)\sinh 2x$

$\frac{dy}{dx} = (x+1) \times 2\cosh 2x + 1 \times \sinh 2x$

$= 2(x+1)\cosh 2x + \sinh 2x$

27) f)

$y = (1-x^2)\cosh\frac{x}{2}$

$\frac{dy}{dx} = (1-x^2) \times \frac{1}{2}\sinh\frac{x}{2} + (-2x) \times \cosh\frac{x}{2}$

$= \frac{1}{2}(1-x^2)\sinh\frac{x}{2} - 2x\cosh\frac{x}{2}$

28)

$y = \sinh 2x \tanh 3x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \sinh 2x \times 3\,\mathrm{sech}^2\, 3x + 2\cosh 2x \times \tanh 3x$

$= 3\sinh 2x\,\mathrm{sech}^2\, 3x + 2\cosh 2x \tanh 3x$

$\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{x=\ln 2} = 3\sinh(2\ln 2)\,\mathrm{sech}^2(3\ln 2) + 2\cosh(2\ln 2)\tanh(3\ln 2)$

$= 3\sinh(\ln 2^2)\,\mathrm{sech}^2(\ln 2^3) + 2\cosh(\ln 2^2)\tanh(\ln 2^3)$

$= 3\sinh(\ln 4)\,\mathrm{sech}^2(\ln 8) + 2\cosh(\ln 4)\tanh(\ln 8)$

$= 3 \times \frac{\mathrm{e}^{\ln 4} - \mathrm{e}^{-\ln 4}}{2} \times \frac{1}{\left(\frac{\mathrm{e}^{\ln 8} + \mathrm{e}^{-\ln 8}}{2}\right)^2} + 2 \times \frac{\mathrm{e}^{\ln 4} + \mathrm{e}^{-\ln 4}}{2} \times \frac{\mathrm{e}^{\ln 8} - \mathrm{e}^{-\ln 8}}{\mathrm{e}^{\ln 8} + \mathrm{e}^{-\ln 8}}$

$= 3 \times \frac{\mathrm{e}^{\ln 4} - \mathrm{e}^{\ln 4^{-1}}}{2} \times \frac{1}{\left(\frac{\mathrm{e}^{\ln 8} + \mathrm{e}^{\ln 8^{-1}}}{2}\right)^2} + 2 \times \frac{\mathrm{e}^{\ln 4} + \mathrm{e}^{\ln 4^{-1}}}{2} \times \frac{\mathrm{e}^{\ln 8} - \mathrm{e}^{\ln 8^{-1}}}{\mathrm{e}^{\ln 8} + \mathrm{e}^{\ln 8^{-1}}}$

$= 3 \times \frac{\mathrm{e}^{\ln 4} - \mathrm{e}^{\ln \frac{1}{4}}}{2} \times \frac{1}{\left(\frac{\mathrm{e}^{\ln 8} + \mathrm{e}^{\ln \frac{1}{8}}}{2}\right)^2} + 2 \times \frac{\mathrm{e}^{\ln 4} + \mathrm{e}^{\ln \frac{1}{4}}}{2} \times \frac{\mathrm{e}^{\ln 8} - \mathrm{e}^{\ln \frac{1}{8}}}{\mathrm{e}^{\ln 8} + \mathrm{e}^{\ln \frac{1}{8}}}$

$= 3 \times \frac{4 - \frac{1}{4}}{2} \times \frac{1}{\left(\frac{8 + \frac{1}{8}}{2}\right)^2} + 2 \times \frac{4 + \frac{1}{4}}{2} \times \frac{8 - \frac{1}{8}}{8 + \frac{1}{8}}$

$= 3 \times \frac{15}{8} \times \frac{256}{4225} + 2 \times \frac{17}{8} \times \frac{63}{65}$

$= \frac{288}{845} + \frac{1071}{260} = \frac{3015}{676}$

29)

$y = (x^2 + x - 6)\cosh x$

$y|_{x=0} = (0^2 + 0 - 6)\cosh 0$

$= -6 \times 1 = -6$

$\frac{\mathrm{d}y}{\mathrm{d}x} = (x^2 + x - 6)\sinh x + (2x + 1)\cosh x$

$\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{x=0} = (0^2 + 0 - 6)\sinh 0 + (0 + 1)\cosh 0$

$= -6 \times 0 + 1 \times 1 = 1$

So the tangent will have equation $y + 6 = x - 0$

$\Rightarrow y = x - 6$

30)

$y = \sinh(12x - \ln 2)\cosh(3x - \ln 3)$

$y|_{x=0} = \sinh(0 - \ln 2)\cosh(0 - \ln 3)$

$= \sinh(\ln 2^{-1})\cosh(\ln 3^{-1})$

$= \sinh\left(\ln\frac{1}{2}\right)\cosh\left(\ln\frac{1}{3}\right)$

$= \frac{e^{\ln\frac{1}{2}} - e^{-\ln\frac{1}{2}}}{2} \times \frac{e^{\ln\frac{1}{3}} + e^{-\ln\frac{1}{3}}}{2}$

$= \frac{e^{\ln\frac{1}{2}} - e^{\ln\left(\frac{1}{2}\right)^{-1}}}{2} \times \frac{e^{\ln\frac{1}{3}} + e^{\ln\left(\frac{1}{3}\right)^{-1}}}{2}$

$= \frac{e^{\ln\frac{1}{2}} - e^{\ln 2}}{2} \times \frac{e^{\ln\frac{1}{3}} + e^{\ln 3}}{2}$

$= \frac{\frac{1}{2} - 2}{2} \times \frac{\frac{1}{3} + 3}{2} = -\frac{5}{4}$

$\frac{dy}{dx} = \sinh(12x - \ln 2) \times 3\sinh(3x - \ln 3) + 12\cosh(12x - \ln 2) \times \cosh(3x - \ln 3)$

$= 3\sinh(12x - \ln 2)\sinh(3x - \ln 3) + 12\cosh(12x - \ln 2)\cosh(3x - \ln 3)$

$\left.\frac{dy}{dx}\right|_{x=0} = 3\sinh(0 - \ln 2)\sinh(0 - \ln 3) + 12\cosh(0 - \ln 2)\cosh(0 - \ln 3)$

$= 3\sinh(\ln 2^{-1})\sinh(\ln 3^{-1}) + 12\cosh(\ln 2^{-1})\cosh(\ln 3^{-1})$

$= 3\sinh\left(\ln\frac{1}{2}\right)\sinh\left(\ln\frac{1}{3}\right) + 12\cosh\left(\ln\frac{1}{2}\right)\cosh\left(\ln\frac{1}{3}\right)$

$= 3 \times \frac{e^{\ln\frac{1}{2}} - e^{-\ln\frac{1}{2}}}{2} \times \frac{e^{\ln\frac{1}{3}} - e^{-\ln\frac{1}{3}}}{2} + 12 \times \frac{e^{\ln\frac{1}{2}} + e^{-\ln\frac{1}{2}}}{2} \times \frac{e^{\ln\frac{1}{3}} + e^{-\ln\frac{1}{3}}}{2}$

$= 3 \times \frac{e^{\ln\frac{1}{2}} - e^{\ln\left(\frac{1}{2}\right)^{-1}}}{2} \times \frac{e^{\ln\frac{1}{3}} - e^{\ln\left(\frac{1}{3}\right)^{-1}}}{2} + 12 \times \frac{e^{\ln\frac{1}{2}} + e^{\ln\left(\frac{1}{2}\right)^{-1}}}{2} \times \frac{e^{\ln\frac{1}{3}} + e^{\ln\left(\frac{1}{3}\right)^{-1}}}{2}$

$= 3 \times \frac{e^{\ln\frac{1}{2}} - e^{\ln 2}}{2} \times \frac{e^{\ln\frac{1}{3}} - e^{\ln 3}}{2} + 12 \times \frac{e^{\ln\frac{1}{2}} + e^{\ln 2}}{2} \times \frac{e^{\ln\frac{1}{3}} + e^{\ln 3}}{2}$

$= 3 \times \frac{\frac{1}{2} - 2}{2} \times \frac{\frac{1}{3} - 3}{2} + 12 \times \frac{\frac{1}{2} + 2}{2} \times \frac{\frac{1}{3} + 3}{2}$

$= 3 + 25 = 28$

So the normal will have equation $y + \frac{5}{4} = -\frac{1}{28}(x - 0)$

$\Rightarrow 28y + 35 = -x$

$\Rightarrow x + 28y = -35$

31)

$y = \sinh x \cosh x$

$\frac{dy}{dx} = \sinh x \times \sinh x + \cosh x \times \cosh x$

$= \sinh^2 x + \cosh^2 x$

$\sinh^2 x \geq 0$ for all real values of x

$\cosh^2 x \geq 1$ for all real values of x

So $\sinh^2 x + \cosh^2 x \geq 1$ for all real values of x

$\therefore \frac{dy}{dx} > 0$ for all x, and hence the curve is always increasing.

32)

$y = (\cosh x - 3)\cosh x$

$\frac{dy}{dx} = (\cosh x - 3)\sinh x + \sinh x \cosh x$

Stationary points exist when $\frac{dy}{dx} = 0$

$\therefore (\cosh x - 3)\sinh x + \sinh x \cosh x = 0$

$\Rightarrow (\cosh x - 3 + \cosh x)\sinh x = 0$

$\Rightarrow (2\cosh x - 3)\sinh x = 0$

$\Rightarrow \cosh x = \frac{3}{2}$ or $\sinh x = 0$

If $\cosh x = \frac{3}{2}$, then $x = \pm \ln\left(\frac{3}{2} + \sqrt{\left(\frac{3}{2}\right)^2 - 1}\right) = \pm \ln\left(\frac{3}{2} + \frac{\sqrt{5}}{2}\right)$

If $\sinh x = 0$, then $x = 0$

$y|_{x=\ln\left(\frac{3}{2}+\frac{\sqrt{5}}{2}\right)} = \left(\cosh\left(\ln\left(\frac{3}{2} + \frac{\sqrt{5}}{2}\right)\right) - 3\right)\cosh\left(\ln\left(\frac{3}{2} + \frac{\sqrt{5}}{2}\right)\right)$

$= \left(\frac{e^{\ln\left(\frac{3}{2}+\frac{\sqrt{5}}{2}\right)} + e^{-\ln\left(\frac{3}{2}+\frac{\sqrt{5}}{2}\right)}}{2} - 3\right) \times \frac{e^{\ln\left(\frac{3}{2}+\frac{\sqrt{5}}{2}\right)} + e^{-\ln\left(\frac{3}{2}+\frac{\sqrt{5}}{2}\right)}}{2}$

$= \left(\frac{e^{\ln\left(\frac{3}{2}+\frac{\sqrt{5}}{2}\right)} + e^{\ln\left(\frac{3}{2}-\frac{\sqrt{5}}{2}\right)}}{2} - 3\right) \times \frac{e^{\ln\left(\frac{3}{2}+\frac{\sqrt{5}}{2}\right)} + e^{\ln\left(\frac{3}{2}-\frac{\sqrt{5}}{2}\right)}}{2}$

$= \left(\frac{\frac{3}{2}+\frac{\sqrt{5}}{2}+\frac{3}{2}-\frac{\sqrt{5}}{2}}{2} - 3\right) \times \frac{\frac{3}{2}+\frac{\sqrt{5}}{2}+\frac{3}{2}-\frac{\sqrt{5}}{2}}{2}$

$= \left(\frac{\frac{3}{2}+\frac{3}{2}}{2} - 3\right) \times \frac{\frac{3}{2}+\frac{3}{2}}{2} = -\frac{9}{4}$

$y|_{x=\ln\left(\frac{3}{2}-\frac{\sqrt{5}}{2}\right)} = \left(\cosh\left(\ln\left(\frac{3}{2} - \frac{\sqrt{5}}{2}\right)\right) - 3\right)\cosh\left(\ln\left(\frac{3}{2} - \frac{\sqrt{5}}{2}\right)\right)$

$$= \left(\frac{e^{\ln\left(\frac{3}{2}-\frac{\sqrt{5}}{2}\right)} + e^{-\ln\left(\frac{3}{2}-\frac{\sqrt{5}}{2}\right)}}{2} - 3 \right) \times \frac{e^{\ln\left(\frac{3}{2}-\frac{\sqrt{5}}{2}\right)} + e^{-\ln\left(\frac{3}{2}-\frac{\sqrt{5}}{2}\right)}}{2}$$

$$= \left(\frac{e^{\ln\left(\frac{3}{2}-\frac{\sqrt{5}}{2}\right)} + e^{\ln\left(\frac{3}{2}+\frac{\sqrt{5}}{2}\right)}}{2} - 3 \right) \times \frac{e^{\ln\left(\frac{3}{2}-\frac{\sqrt{5}}{2}\right)} + e^{\ln\left(\frac{3}{2}+\frac{\sqrt{5}}{2}\right)}}{2}$$

$$= \left(\frac{\frac{3}{2}-\frac{\sqrt{5}}{2}+\frac{3}{2}+\frac{\sqrt{5}}{2}}{2} - 3 \right) \times \frac{\frac{3}{2}-\frac{\sqrt{5}}{2}+\frac{3}{2}+\frac{\sqrt{5}}{2}}{2}$$

$$= \left(\frac{\frac{3}{2}+\frac{3}{2}}{2} - 3 \right) \times \frac{\frac{3}{2}+\frac{3}{2}}{2} = -\frac{9}{4}$$

$y|_{x=0} = (\cosh 0 - 3)\cosh 0$

$= (1 - 3) \times 1 = -2$

So the stationary points are at $\left(\ln\left(\frac{3}{2}+\frac{\sqrt{5}}{2}\right), -\frac{9}{4}\right)$, $\left(\ln\left(\frac{3}{2}-\frac{\sqrt{5}}{2}\right), -\frac{9}{4}\right)$ and $(0, -2)$

33)

$y = \sinh\frac{x}{2}\left(2 - \sinh\frac{x}{3}\right)$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \sinh\frac{x}{2} \times -\frac{1}{3}\cosh\frac{x}{3} + \frac{1}{2}\cosh\frac{x}{2} \times \left(2 - \sinh\frac{x}{3}\right)$

$= -\frac{1}{3}\sinh\frac{x}{2}\cosh\frac{x}{3} + \frac{1}{2}\cosh\frac{x}{2}\left(2 - \sinh\frac{x}{3}\right)$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = -\frac{1}{3}\sinh\frac{x}{2} \times \frac{1}{3}\sinh\frac{x}{3} - \frac{1}{6}\cosh\frac{x}{2} \times \cosh\frac{x}{3} + \frac{1}{2}\cosh\frac{x}{2} \times -\frac{1}{3}\cosh\frac{x}{3} + \frac{1}{4}\sinh\frac{x}{2} \times \left(2 - \sinh\frac{x}{3}\right)$

$= -\frac{1}{9}\sinh\frac{x}{2}\sinh\frac{x}{3} - \frac{1}{6}\cosh\frac{x}{2}\cosh\frac{x}{3} - \frac{1}{6}\cosh\frac{x}{2}\cosh\frac{x}{3} + \frac{1}{4}\sinh\frac{x}{2}\left(2 - \sinh\frac{x}{3}\right)$

$= -\frac{13}{36}\sinh\frac{x}{2}\sinh\frac{x}{3} - \frac{1}{3}\cosh\frac{x}{2}\cosh\frac{x}{3} + \frac{1}{2}\sinh\frac{x}{2}$

$\left.\frac{\mathrm{d}^2y}{\mathrm{d}x^2}\right|_{x=\ln 64} = -\frac{13}{36}\sinh\left(\frac{1}{2}\ln 64\right)\sinh\left(\frac{1}{3}\ln 64\right) - \frac{1}{3}\cosh\left(\frac{1}{2}\ln 64\right)\cosh\left(\frac{1}{3}\ln 64\right) + \frac{1}{2}\sinh\left(\frac{1}{2}\ln 64\right)$

$= -\frac{13}{36}\sinh\left(\ln 64^{\frac{1}{2}}\right)\sinh\left(\ln 64^{\frac{1}{3}}\right) - \frac{1}{3}\cosh\left(\ln 64^{\frac{1}{2}}\right)\cosh\left(\ln 64^{\frac{1}{3}}\right) + \frac{1}{2}\sinh\left(\ln 64^{\frac{1}{2}}\right)$

$= -\frac{13}{36}\sinh(\ln 8)\sinh(\ln 4) - \frac{1}{3}\cosh(\ln 8)\cosh(\ln 4) + \frac{1}{2}\sinh(\ln 8)$

$= -\frac{13}{36} \times \frac{\mathrm{e}^{\ln 8} - \mathrm{e}^{-\ln 8}}{2} \times \frac{\mathrm{e}^{\ln 4} - \mathrm{e}^{-\ln 4}}{2} - \frac{1}{3} \times \frac{\mathrm{e}^{\ln 8} + \mathrm{e}^{-\ln 8}}{2} \times \frac{\mathrm{e}^{\ln 4} + \mathrm{e}^{-\ln 4}}{2} + \frac{1}{2} \times \frac{\mathrm{e}^{\ln 8} - \mathrm{e}^{-\ln 8}}{2}$

$= -\frac{13}{36} \times \frac{\mathrm{e}^{\ln 8} - \mathrm{e}^{\ln 8^{-1}}}{2} \times \frac{\mathrm{e}^{\ln 4} - \mathrm{e}^{\ln 4^{-1}}}{2} - \frac{1}{3} \times \frac{\mathrm{e}^{\ln 8} + \mathrm{e}^{\ln 8^{-1}}}{2} \times \frac{\mathrm{e}^{\ln 4} + \mathrm{e}^{\ln 4^{-1}}}{2} + \frac{1}{2} \times \frac{\mathrm{e}^{\ln 8} - \mathrm{e}^{\ln 8^{-1}}}{2}$

$= -\frac{13}{36} \times \frac{\mathrm{e}^{\ln 8} - \mathrm{e}^{\ln\frac{1}{8}}}{2} \times \frac{\mathrm{e}^{\ln 4} - \mathrm{e}^{\ln\frac{1}{4}}}{2} - \frac{1}{3} \times \frac{\mathrm{e}^{\ln 8} + \mathrm{e}^{\ln\frac{1}{8}}}{2} \times \frac{\mathrm{e}^{\ln 4} + \mathrm{e}^{\ln\frac{1}{4}}}{2} + \frac{1}{2} \times \frac{\mathrm{e}^{\ln 8} - \mathrm{e}^{\ln\frac{1}{8}}}{2}$

$= -\frac{13}{36} \times \frac{8 - \frac{1}{8}}{2} \times \frac{4 - \frac{1}{4}}{2} - \frac{1}{3} \times \frac{8 + \frac{1}{8}}{2} \times \frac{4 + \frac{1}{4}}{2} + \frac{1}{2} \times \frac{8 - \frac{1}{8}}{2}$

$= -\frac{1365}{512} - \frac{1105}{384} + \frac{63}{32} = -\frac{5491}{1536} < 0$

As $\left.\frac{\mathrm{d}^2y}{\mathrm{d}x^2}\right|_{x=\ln 64} < 0$ the curve is concave (concave downwards) at $x = \ln 64$

34)

$y = (\sinh x - 6)\sinh x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = (\sinh x - 6)\cosh x + \cosh x \sinh x$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = (\sinh x - 6)\sinh x + \cosh^2 x + \cosh^2 x + \sinh^2 x$

$= 2\sinh^2 x + 2\cosh^2 x - 6\sinh x$

Points of inflection exist when $\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 0$

$\therefore 2\sinh^2 x + 2\cosh^2 x - 6\sinh x = 0$

$\Rightarrow \sinh^2 x + \cosh^2 x - 3\sinh x = 0$

$\Rightarrow \sinh^2 x + (\sinh^2 x + 1) - 3\sinh x = 0$

$\Rightarrow 2\sinh^2 x - 3\sinh x + 1 = 0$

$\Rightarrow (2\sinh x - 1)(\sinh x - 1) = 0$

$\Rightarrow \sinh x = \frac{1}{2}$ or $\sinh x = 1$

If $\sinh x = \frac{1}{2}$, then $x = \ln\left(\frac{1}{2} + \sqrt{\left(\frac{1}{2}\right)^2 + 1}\right) = \ln\left(\frac{1}{2} + \frac{\sqrt{5}}{2}\right)$

If $\sinh x = 1$, then $x = \ln\left(1 + \sqrt{1^2 + 1}\right) = \ln\left(1 + \sqrt{2}\right)$

$y|_{x=\ln\left(\frac{1}{2}+\frac{\sqrt{5}}{2}\right)} = \left(\sinh\left(\ln\left(\frac{1}{2} + \frac{\sqrt{5}}{2}\right)\right) - 6\right)\sinh\left(\ln\left(\frac{1}{2} + \frac{\sqrt{5}}{2}\right)\right)$

$= \left(\frac{\mathrm{e}^{\ln\left(\frac{1}{2}+\frac{\sqrt{5}}{2}\right)} - \mathrm{e}^{-\ln\left(\frac{1}{2}+\frac{\sqrt{5}}{2}\right)}}{2} - 6\right) \times \frac{\mathrm{e}^{\ln\left(\frac{1}{2}+\frac{\sqrt{5}}{2}\right)} - \mathrm{e}^{-\ln\left(\frac{1}{2}+\frac{\sqrt{5}}{2}\right)}}{2}$

$= \left(\frac{\mathrm{e}^{\ln\left(\frac{1}{2}+\frac{\sqrt{5}}{2}\right)} - \mathrm{e}^{\ln\left(-\frac{1}{2}+\frac{\sqrt{5}}{2}\right)}}{2} - 6\right) \times \frac{\mathrm{e}^{\ln\left(\frac{1}{2}+\frac{\sqrt{5}}{2}\right)} - \mathrm{e}^{\ln\left(-\frac{1}{2}+\frac{\sqrt{5}}{2}\right)}}{2}$

$= \left(\frac{\frac{1}{2}+\frac{\sqrt{5}}{2} - \left(-\frac{1}{2}+\frac{\sqrt{5}}{2}\right)}{2} - 6\right) \times \frac{\frac{1}{2}+\frac{\sqrt{5}}{2} - \left(-\frac{1}{2}+\frac{\sqrt{5}}{2}\right)}{2}$

$= \left(\frac{1}{2} - 6\right) \times \frac{1}{2} = -\frac{11}{4}$

$y|_{x=\ln(1+\sqrt{2})} = \left(\sinh\left(\ln\left(1 + \sqrt{2}\right)\right) - 6\right)\sinh\left(\ln\left(1 + \sqrt{2}\right)\right)$

$= \left(\frac{\mathrm{e}^{\ln(1+\sqrt{2})} - \mathrm{e}^{-\ln(1+\sqrt{2})}}{2} - 6\right) \times \frac{\mathrm{e}^{\ln(1+\sqrt{2})} - \mathrm{e}^{-\ln(1+\sqrt{2})}}{2}$

$= \left(\frac{\mathrm{e}^{\ln(1+\sqrt{2})} - \mathrm{e}^{\ln(-1+\sqrt{2})}}{2} - 6\right) \times \frac{\mathrm{e}^{\ln(1+\sqrt{2})} - \mathrm{e}^{\ln(-1+\sqrt{2})}}{2}$

$= \left(\frac{1+\sqrt{2} - (-1+\sqrt{2})}{2} - 6\right) \times \frac{1+\sqrt{2} - (-1+\sqrt{2})}{2}$

$= (1 - 6) \times 1 = -5$

So we have two candidates: $\left(\ln\left(\frac{1}{2}+\frac{\sqrt{5}}{2}\right),-\frac{11}{4}\right)$ and $\left(\ln\left(1+\sqrt{2}\right),-5\right)$

We need to check that the 2nd derivative changes sign either side of the points we have found. $\ln\left(\frac{1}{2}+\frac{\sqrt{5}}{2}\right)=0.481$ (3 SF) and $\ln\left(1+\sqrt{2}\right)=0.881$ (3 SF)

$\left.\frac{\mathrm{d}^2y}{\mathrm{d}x^2}\right|_{x=0}=2\sinh^2 0+2\cosh^2 0-6\sinh 0=2>0$ (convex (concave upwards))

$\left.\frac{\mathrm{d}^2y}{\mathrm{d}x^2}\right|_{x=0.5}=2\sinh^2 0.5+2\cosh^2 0.5-6\sinh 0.5=-0.0404\ldots<0$ (concave (concave downwards))

$\left.\frac{\mathrm{d}^2y}{\mathrm{d}x^2}\right|_{x=1}=2\sinh^2 1+2\cosh^2 1-6\sinh 1=0.473\ldots>0$ (convex (concave upwards))

$\therefore \left(\ln\left(\frac{1}{2}+\frac{\sqrt{5}}{2}\right),-\frac{11}{4}\right)$ and $\left(\ln\left(1+\sqrt{2}\right),-5\right)$ are both points of inflection for the curve.

35)

$\frac{\mathrm{d}}{\mathrm{d}x}(\tanh x)$

$=\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\sinh x}{\cosh x}\right)$

$=\frac{\cosh x\times\cosh x-\sinh x\times\sinh x}{\cosh^2 x}$

$=\frac{\cosh^2 x-\sinh^2 x}{\cosh^2 x}$

$=\frac{1}{\cosh^2 x}$

$=\operatorname{sech}^2 x$

36) a)

$y=\frac{\sinh x}{\cosh 2x}$

$\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{\cosh 2x\times\cosh x-\sinh x\times 2\sinh 2x}{\cosh^2 2x}$

$=\frac{\cosh x\cosh 2x-2\sinh x\sinh 2x}{\cosh^2 2x}$

36) b)

$y=\frac{\cosh x}{\sinh 2x}$

$\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{\sinh 2x\times\sinh x-\cosh x\times 2\cosh 2x}{\sinh^2 2x}$

$=\frac{\sinh x\sinh 2x-2\cosh x\cosh 2x}{\sinh^2 2x}$

36) c)

$$y = \frac{x}{\sinh x}$$
$$\frac{dy}{dx} = \frac{\sinh x \times 1 - x \times \cosh x}{\sinh^2 x}$$
$$= \frac{\sinh x - x\cosh x}{\sinh^2 x}$$

36) d)

$$y = \frac{\cosh x}{x}$$
$$\frac{dy}{dx} = \frac{x \times \sinh x - \cosh x \times 1}{x^2}$$
$$= \frac{x\sinh x - \cosh x}{x^2}$$

36) e)

$$y = \frac{x+1}{\sinh 2x}$$
$$\frac{dy}{dx} = \frac{\sinh 2x \times 1 - (x+1) \times 2\cosh 2x}{\sinh^2 2x}$$
$$= \frac{\sinh 2x - 2(x+1)\cosh 2x}{\sinh^2 2x}$$

36) f)

$$y = \frac{\cosh 3x}{x^2+1}$$
$$\frac{dy}{dx} = \frac{(x^2+1) \times 3\sinh 3x - \cosh 3x \times 2x}{(x^2+1)^2}$$
$$= \frac{3(x^2+1)\sinh 3x - 2x\cosh 3x}{(x^2+1)^2}$$

37)

$$y = \frac{\sinh 3x}{\sinh x}$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\sinh x \times 3\cosh 3x - \sinh 3x \times \cosh x}{\sinh^2 x}$$

$$= \frac{3\sinh x \cosh 3x - \sinh 3x \cosh x}{\sinh^2 x}$$

$$\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{x=\ln 2} = \frac{3\sinh(\ln 2)\cosh(3\ln 2) - \sinh(3\ln 2)\cosh(\ln 2)}{\sinh^2(\ln 2)}$$

$$= \frac{3\sinh(\ln 2)\cosh(\ln 2^3) - \sinh(\ln 2^3)\cosh(\ln 2)}{\sinh^2(\ln 2)}$$

$$= \frac{3\sinh(\ln 2)\cosh(\ln 8) - \sinh(\ln 8)\cosh(\ln 2)}{\sinh^2(\ln 2)}$$

$$= \frac{3\times\frac{e^{\ln 2}-e^{-\ln 2}}{2}\times\frac{e^{\ln 8}+e^{-\ln 8}}{2} - \frac{e^{\ln 8}-e^{-\ln 8}}{2}\times\frac{e^{\ln 2}+e^{-\ln 2}}{2}}{\left(\frac{e^{\ln 2}-e^{-\ln 2}}{2}\right)^2}$$

$$= \frac{3\times\frac{e^{\ln 2}-e^{\ln 2^{-1}}}{2}\times\frac{e^{\ln 8}+e^{\ln 8^{-1}}}{2} - \frac{e^{\ln 8}-e^{\ln 8^{-1}}}{2}\times\frac{e^{\ln 2}+e^{\ln 2^{-1}}}{2}}{\left(\frac{e^{\ln 2}-e^{\ln 2^{-1}}}{2}\right)^2}$$

$$= \frac{3\times\frac{e^{\ln 2}-e^{\ln\frac{1}{2}}}{2}\times\frac{e^{\ln 8}+e^{\ln\frac{1}{8}}}{2} - \frac{e^{\ln 8}-e^{\ln\frac{1}{8}}}{2}\times\frac{e^{\ln 2}+e^{\ln\frac{1}{2}}}{2}}{\left(\frac{e^{\ln 2}-e^{\ln\frac{1}{2}}}{2}\right)^2}$$

$$= \frac{3\times\frac{2-\frac{1}{2}}{2}\times\frac{8+\frac{1}{8}}{2} - \frac{8-\frac{1}{8}}{2}\times\frac{2+\frac{1}{2}}{2}}{\left(\frac{2-\frac{1}{2}}{2}\right)^2}$$

$$= \frac{\frac{585}{64}-\frac{315}{64}}{\frac{9}{16}} = \frac{15}{2}$$

38)

$$y = \frac{x+\cosh 3x}{1-\sinh 2x}$$

$$y|_{x=0} = \frac{0+\cosh 0}{1-\sinh 0} = \frac{0+1}{1-0} = 1$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{(1-\sinh 2x)\times(1+3\sinh 3x) - (x+\cosh 3x)\times -2\cosh 2x}{(1-\sinh 2x)^2}$$

$$= \frac{(1-\sinh 2x)(1+3\sinh 3x) + 2(x+\cosh 3x)\cosh 2x}{(1-\sinh 2x)^2}$$

$$\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{x=0} = \frac{(1-\sinh 0)(1+3\sinh 0)+2(0+\cosh 0)\cosh 0}{(1-\sinh 0)^2} = \frac{(1-0)(1+0)+2(0+1)\times 1}{(1-0)^2} = 3$$

So the tangent will have equation $y - 1 = 3(x - 0)$

$\Rightarrow y = 3x + 1$

39)

$y = \frac{\tanh 3x+6}{\cosh 2x+3}$

$y|_{x=0} = \frac{\tanh 0+6}{\cosh 0+3} = \frac{0+6}{1+3} = \frac{3}{2}$

$\frac{dy}{dx} = \frac{(\cosh 2x+3)\times 3\,\mathrm{sech}^2\, 3x-(\tanh 3x+6)\times 2\sinh 2x}{(\cosh 2x+3)^2}$

$= \frac{3(\cosh 2x+3)\,\mathrm{sech}^2\, 3x-2(\tanh 3x+6)\sinh 2x}{(\cosh 2x+3)^2}$

$\left.\frac{dy}{dx}\right|_{x=0} = \frac{3(\cosh 0+3)\,\mathrm{sech}^2\, 0-2(\tanh 0+6)\sinh 0}{(\cosh 0+3)^2} = \frac{3(1+3)\times 1-2(0+6)\times 0}{(1+3)^2} = \frac{3}{4}$

So the normal will have equation $y-\frac{3}{2} = -\frac{4}{3}(x-0)$

$\Rightarrow 6y-9 = -8x \Rightarrow 8x+6y = 9$

40) a)

$\mathrm{f}(x) = \frac{x+1}{\sinh x}$

$\mathrm{f}'(x) = \frac{\sinh x\times 1-(x+1)\times\cosh x}{\sinh^2 x} = \frac{\sinh x-(x+1)\cosh x}{\sinh^2 x}$

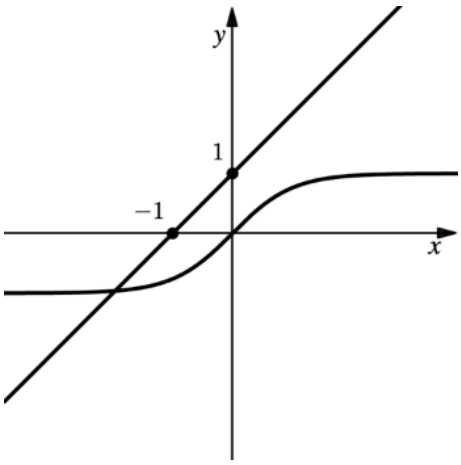
Stationary points exist when $\mathrm{f}'(x) = 0$

$\frac{\sinh x-(x+1)\cosh x}{\sinh^2 x} = 0 \Rightarrow \sinh x-(x+1)\cosh x = 0$

$\Rightarrow \frac{\sinh x}{\cosh x}-(x+1) = 0$

$\Rightarrow \tanh x = x+1$ as required

40) b)



There is one point of intersection between $y = \tanh x$ and $y = x+1$, so the curve $y = \mathrm{f}(x)$ has only one stationary point.

41)

$y = \frac{\sinh x+1}{\tanh x}$

$\frac{dy}{dx} = \frac{\tanh x \times \cosh x - (\sinh x+1) \times \operatorname{sech}^2 x}{\tanh^2 x}$

$= \frac{\frac{\sinh x}{\cosh x} \times \cosh x - (\sinh x+1) \times \frac{1}{\cosh^2 x}}{\tanh^2 x}$

$= \frac{\sinh x - \frac{\sinh x+1}{\cosh^2 x}}{\tanh^2 x}$

Stationary points exist when $\frac{dy}{dx} = 0$

$\therefore \frac{\sinh x - \frac{\sinh x+1}{\cosh^2 x}}{\tanh^2 x} = 0$

$\Rightarrow \sinh x - \frac{\sinh x+1}{\cosh^2 x} = 0$

$\Rightarrow \sinh x \cosh^2 x - \sinh x - 1 = 0$

$\Rightarrow \sinh x (\sinh^2 x + 1) - \sinh x - 1 = 0$

$\Rightarrow \sinh^3 x + \sinh x - \sinh x - 1 = 0$

$\Rightarrow \sinh^3 x - 1 = 0$

$\Rightarrow \sinh^3 x = 1$

$\Rightarrow \sinh x = 1$

$\Rightarrow x = \ln\left(1 + \sqrt{1^2 + 1}\right) = \ln\left(1 + \sqrt{2}\right)$

$y|_{x=\ln(1+\sqrt{2})} = \frac{\sinh(\ln(1+\sqrt{2}))+1}{\tanh(\ln(1+\sqrt{2}))}$

$= \frac{\frac{e^{\ln(1+\sqrt{2})} - e^{-\ln(1+\sqrt{2})}}{2} + 1}{\frac{e^{\ln(1+\sqrt{2})} - e^{-\ln(1+\sqrt{2})}}{e^{\ln(1+\sqrt{2})} + e^{-\ln(1+\sqrt{2})}}}$

$= \frac{\frac{e^{\ln(1+\sqrt{2})} - e^{\ln(-1+\sqrt{2})}}{2} + 1}{\frac{e^{\ln(1+\sqrt{2})} - e^{\ln(-1+\sqrt{2})}}{e^{\ln(1+\sqrt{2})} + e^{\ln(-1+\sqrt{2})}}}$

$= \frac{\frac{1+\sqrt{2}-(-1+\sqrt{2})}{2} + 1}{\frac{1+\sqrt{2}-(-1+\sqrt{2})}{1+\sqrt{2}+(-1+\sqrt{2})}}$

$= \frac{1+1}{\frac{2}{2\sqrt{2}}}$

$= 2\sqrt{2}$

So the stationary point is at $\left(\ln\left(1 + \sqrt{2}\right), 2\sqrt{2}\right)$

42)

$y = \frac{\cosh x}{4-\sinh x}$

$\frac{dy}{dx} = \frac{(4-\sinh x)\times\sinh x-\cosh x\times(-\cosh x)}{(4-\sinh x)^2}$

$= \frac{4\sinh x-\sinh^2 x+\cosh^2 x}{(4-\sinh x)^2}$

$= \frac{4\sinh x+1}{(4-\sinh x)^2}$

$\frac{dy}{dx} = \frac{1}{16}$

$\Rightarrow \frac{4\sinh x+1}{(4-\sinh x)^2} = \frac{1}{16}$

$\Rightarrow 16(4\sinh x + 1) = (4 - \sinh x)^2$

$\Rightarrow 64\sinh x + 16 = 16 - 8\sinh x + \sinh^2 x$

$\Rightarrow 0 = \sinh^2 x - 72\sinh x$

$\Rightarrow 0 = \sinh x\,(\sinh x - 72)$

$\Rightarrow \sinh x = 0$ or $\sinh x = 72$

If $\sinh x = 72$, then $x = \ln\left(72 + \sqrt{72^2 + 1}\right) = \ln\left(72 + \sqrt{5185}\right)$

43)

$y = \frac{4}{\cosh x}$

$\frac{dy}{dx} = \frac{\cosh x \times 0 - 4 \times \sinh x}{\cosh^2 x} = -\frac{4 \sinh x}{\cosh^2 x}$

$\frac{d^2y}{dx^2} = -\frac{\cosh^2 x \times 4 \cosh x - 4 \sinh x \times 2 \sinh x \cosh x}{\cosh^4 x}$

$= -\frac{4 \cosh^3 x - 8 \sinh^2 x \cosh x}{\cosh^4 x}$

$= \frac{8 \sinh^2 x - 4 \cosh^2 x}{\cosh^3 x}$

Points of inflection exist when $\frac{d^2y}{dx^2} = 0$

$\therefore \frac{8 \sinh^2 x - 4 \cosh^2 x}{\cosh^3 x} = 0$

$\Rightarrow 8 \sinh^2 x - 4 \cosh^2 x = 0$

$\Rightarrow \frac{2 \sinh^2 x}{\cosh^2 x} - 1 = 0$

$\Rightarrow 2 \tanh^2 x - 1 = 0$

$\Rightarrow \tanh^2 x = \frac{1}{2}$

$\Rightarrow \tanh x = \frac{1}{\sqrt{2}}$ or $\tanh x = -\frac{1}{\sqrt{2}}$

If $\tanh x = \frac{1}{\sqrt{2}}$, then $x = \frac{1}{2} \ln \left(\frac{1 + \frac{1}{\sqrt{2}}}{1 - \frac{1}{\sqrt{2}}} \right) = \frac{1}{2} \ln \left(\frac{\sqrt{2}+1}{\sqrt{2}-1} \right) = \frac{1}{2} \ln (3 + 2\sqrt{2})$

$= \ln \left((3 + 2\sqrt{2})^{\frac{1}{2}} \right) = \ln (1 + \sqrt{2})$

If $\tanh x = -\frac{1}{\sqrt{2}}$, then $x = \frac{1}{2} \ln \left(\frac{1 + \left(-\frac{1}{\sqrt{2}}\right)}{1 - \left(-\frac{1}{\sqrt{2}}\right)} \right) = \frac{1}{2} \ln \left(\frac{\sqrt{2}-1}{\sqrt{2}+1} \right) = \frac{1}{2} \ln (3 - 2\sqrt{2})$

$= \ln \left((3 - 2\sqrt{2})^{\frac{1}{2}} \right) = \ln (-1 + \sqrt{2})$

$y|_{x = \ln(1+\sqrt{2})} = \frac{4}{\cosh(\ln(1+\sqrt{2}))}$

$= \frac{4}{\frac{e^{\ln(1+\sqrt{2})} + e^{-\ln(1+\sqrt{2})}}{2}}$

$= \frac{8}{e^{\ln(1+\sqrt{2})} + e^{-\ln(1+\sqrt{2})}}$

$= \frac{8}{e^{\ln(1+\sqrt{2})} + e^{\ln\left((1+\sqrt{2})^{-1}\right)}}$

$= \frac{8}{e^{\ln(1+\sqrt{2})} + e^{\ln(-1+\sqrt{2})}}$

$= \frac{8}{(1+\sqrt{2}) + (-1+\sqrt{2})}$

$= \frac{8}{2\sqrt{2}} = 2\sqrt{2}$

$y|_{x=\ln(-1+\sqrt{2})} = \frac{4}{\cosh(\ln(-1+\sqrt{2}))}$

$= \frac{4}{\frac{e^{\ln(-1+\sqrt{2})}+e^{-\ln(-1+\sqrt{2})}}{2}}$

$= \frac{8}{e^{\ln(-1+\sqrt{2})}+e^{-\ln(-1+\sqrt{2})}}$

$= \frac{8}{e^{\ln(-1+\sqrt{2})}+e^{\ln\left((-1+\sqrt{2})^{-1}\right)}}$

$= \frac{8}{e^{\ln(-1+\sqrt{2})}+e^{\ln(1+\sqrt{2})}}$

$= \frac{8}{(-1+\sqrt{2})+(1+\sqrt{2})}$

$= \frac{8}{2\sqrt{2}} = 2\sqrt{2}$

So we have two candidates: $\left(\ln\left(1+\sqrt{2}\right), 2\sqrt{2}\right)$ and $\left(\ln\left(-1+\sqrt{2}\right), 2\sqrt{2}\right)$

We need to check that the 2nd derivative changes sign either side of the points we have found. $\ln\left(1+\sqrt{2}\right) = 0.881$ (3 SF) and $\ln\left(-1+\sqrt{2}\right) = -0.881$ (3 SF)

$\left.\frac{d^2y}{dx^2}\right|_{x=-1} = \frac{8\sinh^2(-1)-4\cosh^2(-1)}{\cosh^3(-1)} = 0.414\ldots > 0$ (convex (concave upwards))

$\left.\frac{d^2y}{dx^2}\right|_{x=0} = \frac{8\sinh^2 0-4\cosh^2 0}{\cosh^3 0} = -4 < 0$ (concave (concave downwards))

$\left.\frac{d^2y}{dx^2}\right|_{x=1} = \frac{8\sinh^2 1-4\cosh^2 1}{\cosh^3 1} = 0.414\ldots > 0$ (convex (concave upwards))

$\therefore \left(\ln\left(1+\sqrt{2}\right), 2\sqrt{2}\right)$ and $\left(\ln\left(-1+\sqrt{2}\right), 2\sqrt{2}\right)$ are both points of inflection for the curve.

44)

$\frac{d}{dx}(\coth x)$

$= \frac{d}{dx}\left(\frac{\cosh x}{\sinh x}\right)$

$= \frac{\sinh x \times \sinh x - \cosh x \times \cosh x}{\sinh^2 x}$

$= \frac{\sinh^2 x - \cosh^2 x}{\sinh^2 x}$

$= \frac{-(\cosh^2 x - \sinh^2 x)}{\sinh^2 x}$

$= -\frac{1}{\sinh^2 x}$

$= -\operatorname{cosech}^2 x$

45)

$$\frac{\mathrm{d}}{\mathrm{d}x}(\operatorname{cosech} x)$$
$$= \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{\sinh x}\right)$$
$$= \frac{\sinh x \times 0 - 1 \times \cosh x}{\sinh^2 x}$$
$$= -\frac{\cosh x}{\sinh^2 x}$$
$$= -\frac{\cosh x}{\sinh x} \times \frac{1}{\sinh x}$$
$$= -\coth x \operatorname{cosech} x$$

46)

$$\frac{\mathrm{d}}{\mathrm{d}x}(\operatorname{sech} x)$$
$$= \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{\cosh x}\right)$$
$$= \frac{\cosh x \times 0 - 1 \times \sinh x}{\cosh^2 x}$$
$$= -\frac{\sinh x}{\cosh^2 x}$$
$$= -\frac{\sinh x}{\cosh x} \times \frac{1}{\cosh x}$$
$$= -\tanh x \operatorname{sech} x$$

47)

$y = 2x + \operatorname{sech} x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 2 - \tanh x \operatorname{sech} x$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = -\tanh x \times -\tanh x \operatorname{sech} x + \operatorname{sech} x \times -\operatorname{sech}^2 x$

$= \tanh^2 x \operatorname{sech} x - \operatorname{sech}^3 x$

$= \operatorname{sech} x (\tanh^2 x - \operatorname{sech}^2 x)$

Points of inflection exist when $\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 0$

$\Rightarrow \operatorname{sech} x (\tanh^2 x - \operatorname{sech}^2 x) = 0$

$\Rightarrow \tanh^2 x - \operatorname{sech}^2 x = 0$ as $\operatorname{sech} x \neq 0$

$\Rightarrow \sinh^2 x - 1 = 0$

$\Rightarrow \sinh^2 x = 1$

$\Rightarrow \sinh x = 1$ or $\sinh x = -1$

If $\sinh x = 1$ then $x = \ln\left(1 + \sqrt{1^2 + 1}\right) = \ln\left(1 + \sqrt{2}\right) \approx 0.881$

If $\sinh x = -1$ then $x = \ln\left(-1 + \sqrt{(-1)^2 + 1}\right) = \ln\left(-1 + \sqrt{2}\right) \approx -0.881$

$\left.\frac{\mathrm{d}^2y}{\mathrm{d}x^2}\right|_{x=0.8} = -0.0883 < 0$ (3SF)

$\left.\frac{\mathrm{d}^2y}{\mathrm{d}x^2}\right|_{x=0.9} = 0.0183 > 0$ (3SF)

$\left.\frac{\mathrm{d}^2y}{\mathrm{d}x^2}\right|_{x=-0.9} = 0.0183 > 0$ (3SF)

$\left.\frac{\mathrm{d}^2y}{\mathrm{d}x^2}\right|_{x=-0.8} = -0.0883 < 0$ (3SF)

As the 2nd derivative changes sign either side of the two points, there must be points of inflection at $x = \ln\left(1 + \sqrt{2}\right)$ and at $x = \ln\left(-1 + \sqrt{2}\right)$

48)

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\ln\tanh\frac{x}{2}\right)$$

$$=\frac{\frac{1}{2}\operatorname{sech}^2\frac{x}{2}}{\tanh\frac{x}{2}}$$

$$=\frac{1-\tanh^2\frac{x}{2}}{2\tanh\frac{x}{2}}$$

$$=\frac{1-\frac{\sinh^2\frac{x}{2}}{\cosh^2\frac{x}{2}}}{\frac{2\sinh\frac{x}{2}}{\cosh\frac{x}{2}}}$$

$$=\frac{\cosh^2\frac{x}{2}-\sinh^2\frac{x}{2}}{2\sinh\frac{x}{2}\cosh\frac{x}{2}}$$

$$=\frac{1}{2\sinh\frac{x}{2}\cosh\frac{x}{2}}$$

$$=\frac{1}{\sinh\left(2\times\frac{x}{2}\right)}$$

$$=\frac{1}{\sinh x}$$

$$=\operatorname{cosech} x$$

49)

$y=\arctan(4\mathrm{e}^{5x})$

$\frac{\mathrm{d}y}{\mathrm{d}x}=20\mathrm{e}^{5x}\times\frac{1}{(4\mathrm{e}^{5x})^2+1}=\frac{20\mathrm{e}^{5x}}{16\mathrm{e}^{10x}+1}=\frac{20}{16\mathrm{e}^{5x}+\mathrm{e}^{-5x}}$

$p\sinh 5x+q\cosh 5x\equiv p\left(\frac{\mathrm{e}^{5x}-\mathrm{e}^{-5x}}{2}\right)+q\left(\frac{\mathrm{e}^{5x}+\mathrm{e}^{-5x}}{2}\right)\equiv\frac{1}{2}(p+q)\mathrm{e}^{5x}+\frac{1}{2}(-p+q)\mathrm{e}^{-5x}$

$\frac{1}{2}(p+q)=16\Rightarrow p+q=32$ (1)

$\frac{1}{2}(-p+q)=1\Rightarrow -p+q=2$ (2)

Solve (1) and (2):

$p=15$ and $q=17$

$\therefore\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{20}{15\sinh 5x+17\cosh 5x}$